Traffic Accident Severity Prediction Using Machine Learning and Deep Learning Models

Abstract—**Road traffic accidents are a major public-safety challenge because they produce injuries, fatalities, infrastructure damage, and economic loss. Predicting accident severity before full emergency assessment can support faster response, safer road management, and better municipal planning. This paper presents a complete predictive modeling framework for traffic accident severity classification using classical machine learning and deep learning techniques. The dataset contains 25,791 crash records from 2017 to 2023 with roadway, weather, lighting, temporal, vehicle-count, and crash-cause attributes. Data cleaning, leakage removal, target binarization, feature engineering, categorical one-hot encoding, feature scaling, and train-validation-test splitting were performed before model training. Logistic Regression, Decision Tree, Random Forest, XGBoost, and a Keras-based Deep Neural Network were implemented and evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Experimental results show that Logistic Regression achieved the highest accuracy of 74.00%, while the Deep Neural Network achieved competitive ROC-AUC of 0.8111 and improved recall compared with several baseline models. The findings demonstrate that structured preprocessing and model comparison can produce reliable accident-severity prediction systems for intelligent transportation and traffic-safety applications.**

Index Terms—**Traffic accident prediction, accident severity, machine learning, deep learning, random forest, XGBoost, neural networks, intelligent transportation systems.**

I. Introduction

Traffic accidents are among the most serious challenges faced by modern transportation systems. Rapid urbanization, increasing vehicle density, changing weather conditions, and complex road networks contribute to the uncertainty and severity of crash events. A collision may lead only to property damage, but it may also involve injury, towing, or emergency medical response. For municipal traffic departments, emergency response teams, and road-safety planners, early prediction of severity can support improved resource allocation and faster decision-making.

Traditional accident analysis often relies on descriptive statistics, such as yearly crash counts, accident hotspots, and aggregate injury rates. Although such approaches are useful for post-event reporting, they are limited in their ability to model complex interactions among environmental, temporal, and roadway variables. Machine learning provides a stronger alternative because it can learn patterns from high-dimensional crash datasets and estimate the probability of severe outcomes.

This project focuses on binary accident severity classification. The original target attribute, *crash_type*, was transformed into two classes: no injury/drive-away cases and injury and/or tow-due-to-crash cases. The main objective is to compare classical machine learning algorithms with a deep neural network and determine which approach provides more reliable predictive performance. The study also emphasizes feature engineering, leakage prevention, model evaluation, and comparative analysis using multiple performance metrics.

The major contributions of this work are as follows:

- Development of a preprocessing pipeline for traffic crash severity prediction using mixed categorical and numerical features.
- Engineering of interpretable temporal features such as day period, season, and weekend indicator.
- Comparative evaluation of Logistic Regression, Decision Tree, Random Forest, XGBoost, and a Deep Neural Network.
- Performance analysis using accuracy, precision, recall, F1-score, ROC-AUC, training time, and prediction time.

II. Methodology

The proposed workflow follows an end-to-end data science pipeline consisting of data loading, cleaning, exploratory analysis, preprocessing, model training, and evaluation. The implementation was carried out in Python using Scikit-learn, XGBoost, TensorFlow, Keras, Pandas, NumPy, Matplotlib, and Seaborn.

A. Dataset Description

The dataset contains 25,791 traffic accident records collected from 2017 to 2023. Each record represents a crash incident and includes multiple categories of information. The selected predictive variables include traffic control device, weather condition, lighting condition, first crash type, trafficway type, alignment, roadway surface condition, road defect, intersection-related indicator, damage category, primary contributory cause, number of vehicles involved, crash hour, crash day of week, and crash month.

The original target variable contains two textual classes. These were encoded into a binary label:

$$y = \begin{cases} 0, & \text{No Injury / Drive Away} \\ 1, & \text{Injury and/or Tow Due to Crash.} \end{cases} \tag{1}$$

The dataset showed 14,577 no-injury/drive-away cases and 11,214 injury/tow cases. This distribution indicates moderate class imbalance, making metrics such as recall, F1-score, and ROC-AUC important in addition to accuracy.

B. Data Cleaning and Leakage Prevention

Accident datasets often include columns that are not suitable for predictive modeling because they reveal the target outcome directly. To avoid data leakage, injury-specific variables such as total injuries, fatal injuries, incapacitating injuries, non-incapacitating injuries, reported-not-evident injuries, no-indication injuries, and most severe injury were removed before training. Missing values were removed after selecting modeling features, resulting in a clean modeling frame.

C. Feature Engineering

Three additional interpretable features were created to enhance the representation of accident context. First, *day period* was generated from the crash hour and grouped into morning rush, midday, evening rush, night, and late night. Second, *season* was derived from crash month and grouped into winter, spring, summer, and fall. Third, *is weekend* was computed from crash day of week to distinguish weekday and weekend crash patterns.

D. Encoding and Scaling

Categorical variables were transformed using one-hot encoding. Numerical variables, including number of units, crash hour, crash day of week, and crash month, were standardized using StandardScaler. The dataset was divided into training, validation, and test subsets using fixed random seeds to ensure reproducibility.

III. Exploratory Data Analysis

Exploratory data analysis was performed to understand accident patterns before modeling. The target distribution showed that no-injury/drive-away crashes were more frequent than injury/tow crashes, but the severe class still represented a substantial portion of the dataset. Hourly analysis indicated that crash frequency varies by time of day, and the engineered day-period feature was designed to capture this temporal effect.

Weather and lighting conditions were analyzed because visibility and roadway conditions can influence both crash likelihood and severity. The notebook also examined first crash type, primary contributory cause, monthly trends, correlation among numerical variables, and number of vehicles involved. These analyses guided feature selection and supported the decision to include both environmental and temporal predictors in the final modeling frame.

IV. Predictive Model Architectures

Five predictive models were implemented to compare linear, tree-based, ensemble, boosting, and neural approaches.

A. Logistic Regression

Logistic Regression was selected as a baseline linear classifier. It models the probability of a positive crash-severity outcome using a logistic sigmoid function. Despite its simplicity, it is useful for determining whether the engineered and encoded features provide linearly separable information.

TABLE I: Architecture Search Results for the Deep Neural Network

Architecture	Hidden Layers	Val. AUC	Val. Acc.
Small	[64, 32]	0.8044	0.7376
Medium	[128, 64, 32]	**0.8072**	0.7360
Large	[256, 128, 64, 32]	0.8061	0.7332
Wide	[256, 128]	0.8057	0.7311

B. Decision Tree

The Decision Tree classifier recursively partitions the feature space using decision rules. It can capture non-linear interactions but may overfit when trained on high-cardinality categorical encodings. Its inclusion provides a simple non-parametric benchmark.

C. Random Forest

Random Forest is an ensemble of decision trees trained using bootstrap aggregation. It reduces variance compared with a single tree and is often effective for structured tabular data. It was included to test whether ensemble averaging improves severity prediction.

D. XGBoost

XGBoost is a gradient boosting algorithm that builds decision trees sequentially, with each new tree correcting errors made by previous learners. It is widely used for tabular classification tasks due to its predictive strength and training efficiency.

E. Deep Neural Network

A deep neural network was developed using TensorFlow and Keras. The model consists of dense layers with ReLU activation, batch normalization, dropout regularization, and L2 weight regularization. Binary cross-entropy was used as the loss function, and Adam was used as the optimizer. Class weights were applied during training to account for class imbalance.

An architecture search was conducted using four candidate designs: small, medium, large, and wide. The medium configuration, consisting of hidden layers [128, 64, 32], achieved the best validation ROC-AUC of 0.8072 and was selected for final training. Early stopping, model checkpointing, and learning-rate reduction were used to improve generalization.

V. Experimental Results and Discussion

The models were evaluated on the final test split. Accuracy, precision, recall, F1-score, ROC-AUC, fitting time, and prediction time were recorded. Table II summarizes the final comparison.

The results indicate that Logistic Regression achieved the highest overall accuracy and ROC-AUC. This suggests that after one-hot encoding and feature engineering, the dataset contains a meaningful linear signal. Random Forest achieved the highest recall and F1-score, making it attractive when the goal is to identify as many severe accidents as possible. Higher recall is important in safety-critical systems because missing a

TABLE II: Final Model Comparison on the Test Set

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Fit Time (s)	Predict Time (s)
Logistic Regression	**0.7400**	**0.7158**	0.6665	0.6903	**0.8115**	1.51	0.0385
Keras Deep Neural Network	0.7374	0.6920	0.7134	0.7026	0.8111	7.33	0.2582
Random Forest	0.7395	0.6794	**0.7586**	**0.7169**	0.8069	2.52	0.0697
XGBoost	0.7351	0.7007	0.6819	0.6912	0.8062	1.31	0.0118
Decision Tree	0.7227	0.6803	0.6831	0.6817	0.7855	**0.14**	**0.0023**

severe accident can have greater consequences than producing a false alarm.

The Keras Deep Neural Network performed competitively, with ROC-AUC close to Logistic Regression and a recall higher than Logistic Regression and XGBoost. This indicates that the neural model learned useful non-linear representations, although its training and prediction times were higher than most classical models. XGBoost achieved balanced performance with fast prediction time, while the Decision Tree produced the lowest ROC-AUC, which is consistent with the tendency of individual trees to overfit tabular data.

Feature importance analysis from XGBoost and Random Forest showed that crash type descriptors and damage-related variables were among the most influential predictors. Examples of high-importance features included pedestrian crash type, damage over $1,500, sideswipe same direction, pedal cyclist crash type, and rear-end crash type. These variables are logically connected to accident severity and support the interpretability of the predictive system.

VI. Conclusion and Future Work

This paper presented a complete IEEE-style predictive modeling study for traffic accident severity classification using machine learning and deep learning models. The workflow included data preprocessing, leakage removal, feature engineering, exploratory analysis, model training, architecture search, and final performance comparison. The dataset contained 25,791 traffic accident records from 2017 to 2023, and the target variable was converted into a binary classification problem.

Experimental results showed that Logistic Regression achieved the highest accuracy of 74.00% and ROC-AUC of 0.8115. Random Forest produced the highest recall of 0.7586 and F1-score of 0.7169, making it particularly suitable for safety-focused applications where detecting severe accidents is critical. The Deep Neural Network also demonstrated competitive performance with ROC-AUC of 0.8111 and recall of 0.7134.

Future work may extend this framework by integrating geographic information systems, real-time traffic sensors, weather feeds, and advanced class-balancing methods such as SMOTE. Additional research may also explore recurrent neural networks, attention-based neural architectures, transformer models, and deployment within intelligent transportation systems for real-time accident severity prediction.

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